

ABSTRACT

**of the PhD thesis by Yesmukhamedov Nurmaganbet Seitkaliuly on
«Computer System for Processing and Analysis of Eye Data to Support
Laser Coagulation in the Treatment of Diabetic Retinopathy», submitted
for the degree of Doctor of Philosophy (PhD) in the educational
programme 8D06102 — Computer and Software Engineering**

General characteristics of the work. The dissertation investigates and formalises the integration of fundus image enhancement (preprocessing) with convolutional-neural-network (CNN) classification for automated multi-stage diagnosis of diabetic retinopathy (DR). Its central idea is that preprocessing is an integral component of the diagnostic model rather than ancillary data preparation, because it defines the feature space available to the network. To operationalise it, an 8-stage preprocessing pipeline is specified and embedded in the model, and its effect is placed under controlled contrast against an equivalent baseline configuration trained without the pipeline, on eight public and clinical fundus datasets.

Relevance of the research. Diabetic retinopathy is a microvascular complication of diabetes and one of the leading causes of preventable vision loss among adults of working age. Its earliest stages, in which microaneurysms and small haemorrhages appear without affecting acuity, are asymptomatic, so a patient has no reason to seek care during the interval in which intervention is most effective. Detection therefore depends not on presentation but on scheduled imaging of a cohort defined by a systemic diagnosis. Screening is, before it is a diagnostic problem, a problem of volume, and volume is what makes the capacity constraint binding: manual grading scales linearly with specialist time, which is finite and geographically concentrated. That automation is technically feasible has been established for a decade, so the relevance of this work rests not on a claim that automated screening remains undemonstrated, but on the conditions under which the demonstrations were obtained: reported performance has been shown to be unevenly robust, and the images are not a stable substrate — photographs acquired by different cameras, operators and dilation states vary systematically, and where screening capacity is scarcest, acquisition conditions are least controlled. The methodological question follows: it concerns how models of this kind are specified rather than how well any one performs. Preprocessing may be treated as ancillary data preparation not requiring methodological discussion, or as an integral component of the model, because the transform applied before the first convolution determines the feature space the network is obliged to operate in. If the second is right, a model reported without its preprocessing specified has not been fully described, and comparisons between such models are comparisons between partly unknown systems.

The aim of the research is to develop and experimentally validate an integrated fundus image enhancement and CNN classification framework for automated five-class DR diagnosis, in which an 8-stage preprocessing pipeline is specified as an integral component of the model, and to establish, under controlled

contrast against an equivalent configuration trained without that pipeline, what difference the specification makes to performance, transferability and interpretability.

Research objectives: 1. To analyse the problem domain and formulate the research problem: the clinical grading of the disease, the screening requirements of resource-limited settings, the sources of image-quality loss and the methodological practice of existing automated systems (Chapter 1). 2. To specify the integrated methodology: the 8-stage pipeline, the classification architectures, the pretraining and fine-tuning strategy, the explainability formalism and the evaluation protocol (Chapter 2). 3. To evaluate the integrated configuration experimentally, in domain and on seven external corpora, and to validate the results statistically (Chapter 3). 4. To build and describe a screening system around the model, suited to settings without inference acceleration and with intermittent connectivity (Chapter 4).

The object of the research is the process of automated multi-stage diagnosis of diabetic retinopathy from colour fundus photographs by means of convolutional networks, studied in its entirety — from the image as it leaves the camera to the grade the model assigns.

The subject of the research is the set of methods by which fundus image preprocessing is integrated with CNN classification, together with the properties the resulting configuration exhibits: its five-class diagnostic performance, its transferability to corpora not seen in training, the alignment of its attention with expert annotation, and its behaviour across camera domains.

Research methods. The methodological framework is controlled experimental comparison, in which configurations are contrasted with everything outside the manipulated factor held fixed. Its instruments are drawn from image processing, deep learning and statistical inference: optic-disc and fovea localisation by a pre-trained, frozen heatmap-regression detector; isotropic rescaling with centred padding made explicit to the network as a mask channel; adaptive illumination correction and dual-constraint CLAHE in the LAB colour space; classification with the ResNet-50 and EfficientNet-B3 architectures under a weighted focal loss; Grad-CAM explainability with the Attention–Lesion Overlap and IoU measures; maximum mean discrepancy over penultimate-layer features; and patient-level stratified 5-fold cross-validation with paired testing, bootstrap intervals and correction for multiplicity.

The empirical basis of the research is a tiered corpus of eight public and clinical fundus datasets from cameras of four manufacturers (Canon, Topcon, Kowa, Zeiss): EyePACS (~35,126 graded images, training and ablation), whose unlabelled split (~53,576 images) supplies the in-domain pretraining corpus and is disjoint from the training set; APTOS 2019 (~3,662, cross-dataset transfer); IDRiD (516 images, 81 with pixel-level lesion masks) and Messidor-2 (~1,748) for external clinical evaluation; DDR (~13,673), ODIR-5K (~5,000) and RFMiD (~3,200) for device-shift evaluation; and a Kazakh clinical dataset (60 images, 30 patients $\times$ 2 eyes) for training in a data-scarce regime. Results are reported per corpus and never aggregated into a single figure of merit.

The scientific novelty of the research consists in the following: 1. The principal contribution is conceptual: preprocessing is reframed from ancillary preparation to an integral component of the diagnostic model — formalised as a

binding part of the model specification and placed under controlled experimental contrast, so that what is new is not the existence of a pipeline but the treatment of one as an object of experiment. 2. Five elements of its engineering realisation are specified in forms not previously combined: isotropic resize with centred padding; a field-of-view mask supplied as a fourth input channel; illumination correction scaled to per-image geometry and applied inside the mask only; channel statistics computed from valid fundus pixels; and canonical orientation whose augmentation dispersion derives from the uncertainty of the landmark localisation. 3. The clip limit of the contrast stage is the minimum of a histogram-relative and a tile-relative constraint, applied stochastically at training time so the stage serves as both enhancement and regularisation; the formulation extends the candidate's previously published work. 4. Attention alignment is measured by the fraction of an annotated lesion that model attention covers, on the reasoning that lesion coverage is the clinically meaningful direction. 5. The postulated mechanism is measured rather than inferred: source-to-target distance is computed at the penultimate layer under both configurations across six external corpora, whereas work of this kind ordinarily judges the mechanism by external accuracy. 6. A defect shared by a family of measures in common use for external robustness is identified analytically: each normalises an arm's external performance against that same arm's in-domain performance and so penalises a configuration for its in-domain strength.

The main results of the research. 1. A critical analysis of the context of automated DR diagnosis and of fundus image quality variability was carried out, the limitations of the end-to-end paradigm were identified and the research problem was formulated. 2. An 8-stage preprocessing pipeline was designed and formalised: canonical flip, OD-fovea rotation normalisation, field-of-view crop with isotropic resize and padding, FOV mask as a fourth input channel, adaptive illumination correction, dual-constraint stochastic CLAHE, augmentation and dataset-specific normalisation. 3. The dominance of the integrated configuration over the baseline was established on the training corpus for both architectures, in all three components of the conjunctive criterion. 4. The contribution of the individual stages was quantified by cumulative ablation under a single initialisation, and the parameters of the contrast stage and of the illumination correction were found to have an interior optimum confirmed on held-out data. 5. The reduction of distance to the training distribution was measured directly across six external corpora under a mandatory source-domain-statistics condition. 6. Transfer to corpora not seen in training was demonstrated — across datasets, across camera groupings and on two external clinical corpora — as was the superiority of the integrated configuration in a data-scarce regime. 7. A modular architecture for an automated DR screening system in resource-limited settings was designed and a working demonstrator of it was built, which runs the pipeline and the trained classifier end to end, produces attention maps and keeps a per-case record.

Main provisions put forward for defence: 1. Preprocessing of fundus images is a formalisable and experimentally testable component of the diagnostic model rather than ancillary preparation of the data; the provision is methodological. 2. The integrated configuration exceeds the baseline on the training corpus of 35,126

images, on both architectures, in all three components of the conjunctive criterion: weighted F1 by 6.54 and 6.55 percentage points, area under the curve by 0.032 and 0.036, and quadratic kappa by 0.11; every interval excludes zero, the comparisons survive correction for multiplicity, and there is no interaction between arm and architecture. The provision concerns the configuration as a whole: the arms differ in initialisation as well, so no part of the effect is attributed to preprocessing alone. 3. The contributions of the individual stages are separable: cumulative ablation over eight levels under one initialisation raises weighted F1 from 0.7538 to 0.8193, monotonically and without a single inversion in any of the five folds, each of the seven transitions contributing 0.0065 to 0.0143 against a between-fold dispersion of 0.0042 to 0.0060, and the two photometric stages together carry 41% of the gain. Both parameters exhibit an interior optimum confirmed on held-out data: a clip factor of 2.5 with a histogram threshold of 0.03, and a correction scale of 0.07 of the field diameter. 4. Distance from the training distribution falls on every one of six external corpora — by 0.070 to 0.093 at the penultimate layer, with every interval excluding zero — achieved without the transform observing any target corpus. 5. Competence transfers to corpora not seen in training: a generalisation ratio of 0.898 against 0.858 and a weighted F1 higher by 0.089 on the cross-corpus set; across five camera groupings performance is higher on every one and the spread between them contracts by factors of 2.4 and 3.1; and on two external clinical corpora the margin over the baseline is 0.069 and 0.054 in weighted F1, above the minimal clinically important difference of 0.050. 6. Model attention overlaps expert-annotated lesions more under the integrated configuration on all four annotated lesion types — by 0.099 to 0.129 at $p \leq 0.0148$ — and the direction holds at every binarisation threshold; the provision is defended as alignment, not as localisation of pathology. 7. Also submitted are a coupled thermal-optical model of fundus tissue response under laser exposure, as a theoretical contribution, and a modular screening-system architecture as a design contribution, realised in part as a working demonstrator.

Theoretical significance lies in how the problem is posed and measured. The reframing changes what counts as a complete description of a diagnostic model of this class, and with it what counts as a fair comparison. Five formalisations make previously informal choices explicit and testable: the clip limit, the illumination correction, attention agreement as an asymmetric overlap, the postulated mechanism as a measurable distance, and the non-neutrality of a family of robustness measures in common use. A mechanism hitherto invoked as an explanation is made measurable — as a quantity computable at the penultimate layer under a stated protocol condition — which makes the mechanistic claim falsifiable independently of the performance claims it explains.

Practical significance. The work makes four things practically available, each with its limit. A fully specified preprocessing regime, reproduced in an appendix, whose parameter values should be re-established rather than inherited. A screening system with preprocessing-engine, inference, store-and-forward telemedicine and physician-in-the-loop decision-support modules, integrable with PACS/EHR systems and the national eHealth platform; a working demonstrator of it is deployed and performs inference on submitted images, while the deployment-oriented parts remain

design specifications and the data-protection provisions are alignment by design rather than certified compliance. A protocol for ingesting externally sourced images, validated only against the clinical source it was built for. And an argument of applicability to the national screening context, bounded by the absence of field testing there. The system is conceived as decision support in which the clinician retains the diagnosis and the responsibility for it.

Reliability of the results rests on procedure rather than on magnitude. Partitioning is patient-level and grade-stratified; performance is reported on a hierarchy of measures fixed in advance; differences are assessed by paired testing on identical cases, their uncertainty quantified by resampling, and a mixed-effects model separates fold variation from the effect under test. The strongest of these grounds is not visible in a table and is therefore stated: the acceptance criterion for each hypothesis was fixed before the experiment that tested it, and a criterion written once a result is known can always be satisfied. Three qualifications bound the apparatus in general: correction for multiplicity is scoped to the single experiment within which the comparisons were planned; several evaluations rest on the models of one fitted fold and so understate total uncertainty in a known direction; and one experiment depends on a clinical corpus that cannot be redistributed. Four attach to particular provisions: the stage ranking holds at the resolution of groupings only, and the two parameter optima are properties of this corpus rather than portable constants; the reduction in distance holds in direction only — its size predicts no gain; neither the generalisation ratio nor the camera-group floor discriminates between the arms, so the transfer evidence lies in the comparison, and on the second external clinical corpus the margin over the minimal important difference is four thousandths; and the attention result rests on one annotated corpus. Not submitted for defence are clinical validity, readiness for unsupervised deployment, device certification, localisation of pathology, and superiority over any named published system.

Approbation of the results and connection with scientific programmes. The results of the research were reported and discussed at the 3rd International Workshop on Digital Society (DS 2025, Istanbul, Türkiye, 28–30 October 2025) and published in international peer-reviewed journals. The direction of the research corresponds to the state priorities of the Republic of Kazakhstan in the digitalisation of healthcare and the development of artificial-intelligence technologies, in particular the Concept for the Development of Artificial Intelligence for 2024–2029, the Address of the President of the Republic of Kazakhstan «Kazakhstan in the Era of Artificial Intelligence» of 8 September 2025, and the Law of the Republic of Kazakhstan «On Artificial Intelligence» of 17 November 2025. The work is carried out in accordance with subparagraph 2 of paragraph 3 of article 20 of the Law of the Republic of Kazakhstan «On Science». This is a correspondence between the direction of the research and published national priorities, not a statement that the work was funded under a state programme or commissioned by any body.

Publications. The main results of the dissertation are published in 5 scientific works, including: 1 article in a journal indexed in Scopus and Web of Science (*Eastern-European Journal of Enterprise Technologies*, Q3, CiteScore 2025 = 2.5, 42nd percentile in the category Computer Science Applications); 1 paper in Scopus-

indexed international conference proceedings (*Procedia Computer Science*); and 3 articles in journals recommended by the Committee for Quality Assurance in Science and Higher Education of the Republic of Kazakhstan.

The results obtained on the topic of the dissertation are presented in the following publications:

1. Sapakova S., Yesmukhamedov N., Sapakov A. Development of an image quality enhancement approach for diabetic retinopathy diagnosis // *Eastern-European Journal of Enterprise Technologies*. — 2025. — Vol. 4, No. 9(136). — P. 79–88. <https://doi.org/10.15587/1729-4061.2025.335570>.

2. Sapakova S., Yesmukhamedov N., Sapakov A., Yemberdiyeva A., Kozhamkulova Z. Methods for pre-processing and analysis of fundus images for detection of diabetic retinopathy // *Procedia Computer Science*. — 2025. — Vol. 272. — P. 496–501 (The 3rd International Workshop on Digital Society, DS 2025, Istanbul, Türkiye). <https://doi.org/10.1016/j.procs.2025.10.237>.

3. Yesmukhamedov N.S., Sapakova S., Al-Haddad S.A.R., Daniyarova D. Development of an information system architecture for healthcare institutions using artificial intelligence // *News of the National Academy of Sciences of the Republic of Kazakhstan, Physico-Mathematical Series*. — 2025. — Vol. 2(354). — P. 74–91. <https://doi.org/10.32014/2025.2518-1726.345>.

4. Yesmukhamedov N.S., Sapakova S.Z., Kozhamkulova Zh.Zh., Daniyarova D.R., Armankyzy R. Methods for preprocessing and analysis of fundus images for diabetic retinopathy detection // *Herald of the Kazakh-British Technical University*. — 2025. — No. 4(75), Vol. 22. — P. 119–130. <https://doi.org/10.55452/1998-6688-2025-22-4-119-130>.

5. Sapakova S.Z., Daniyarova D.R., Yesmukhamedov N.S., Armankyzy R., Yemberdiyeva A.B., Kaldybaeva A.S. Mathematical modeling of laser exposure on fundus tissues in the treatment of diabetic retinopathy // *Herald of KazUTB*. — 2025. — Vol. 2, No. 27-740. — P. 20–30. <https://doi.org/10.58805/kazutb.v.2.27-740>.

Main content of the work. The introduction substantiates the relevance of the topic and formulates the aim, the objectives, the object and subject of the research, the scientific novelty, the provisions submitted for defence, the methodological basis, the significance of the work, the reliability of the results, the approbation and the publications. Chapter 1 establishes the clinical and technical context of screening: it characterises the grading of the disease and what a screening programme demands of a resource-limited setting, identifies the sources of image-quality loss and its device-specific component, reviews what the field has done with convolutional networks and what its automated systems assume about preprocessing, and formulates the research problem that follows from that practice. Chapter 2 specifies the methodology: it sets out the 8-stage pipeline with the theory grounding each stage, formalises the clip limit and the illumination correction, describes the classification architectures and their adaptation to a four-channel input, states the pretraining and fine-tuning strategy and the explainability formalism, and fixes the evaluation protocol in advance. Chapter 3 reports the experimental programme: the in-domain factorial contrast, the stage decomposition with its parameter sweeps, the direct measurement of source-to-target distance, cross-corpus and external clinical transfer, the alignment of attention

with expert annotation, behaviour across camera groupings and in a data-scarce regime, the statistical validation and the limitations bounding the results. Chapter 4 describes the screening system: its architecture, its inference behaviour without acceleration hardware and its ingestion protocol, distinguishing what exists from what remains specification. The conclusion consolidates the outcomes and the directions of further work.

Author's personal contribution. The principal results of the dissertation were obtained by the author personally: the conceptual reframing of preprocessing as an integral component of the diagnostic model; the design and formal specification of the 8-stage pipeline, including the dual-constraint stochastic CLAHE, the adaptive illumination correction and the FOV-mask input channel; the Attention–Lesion Overlap measure; the controlled experimental design and the statistical validation framework; and the architecture of the screening system. All five publications are co-authored; in each the author contributed the part on which this dissertation rests: the formulation of the preprocessing problem, the design and implementation of the pipeline, the conduct of the experiments and the writing of the corresponding sections.

Structure and scope of the work. The dissertation comprises front matter, an introduction, four chapters, a conclusion, a list of references and five appendices. The dissertation is set out on 113 pages and contains 19 tables and 16 figures. The list of references comprises 102 sources. The counts are of the dissertation excluding the appendices.